

K2-29b

R-2194

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_220107 · created 2026-08-14T00:05:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459586.7777 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.043
Transit depth (Rp/Rs)^2	0.97 +/- 0.86 [%]
Orbital Inclination (inc)	89.83 +/- 2.41
Airmass coefficient 1 (a1)	1.022 +/- 0.018
Airmass coefficient 2 (a2)	-0.0115 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	#2 - [167, 415]
Optimal Aperture	3.67
Optimal Annulus	9.19
Transit Duration (day)	0.0509 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.099 ± 0.043 vs 0.1419 (deviation 0.92σ)
Duration fit vs published	0.0509 d vs 0.0925 d (deviation 4.05σ)
Mid-time O–C	12.3 min
Depth SNR	2.1
Residual scatter	1.71 %

Light curve

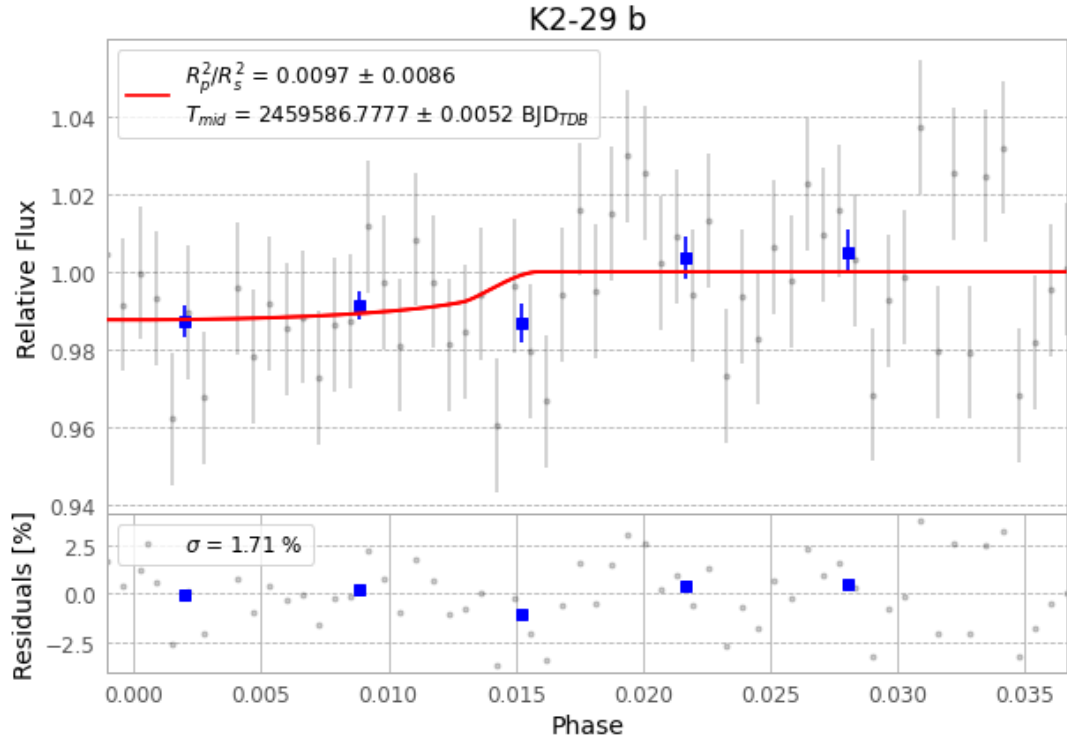


Figure 1 — FinalLightCurve_K2-29 b_2022-01-06.png (SHA-256 4867067bb7d33ba2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 275], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 3412357c6b02d924...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

60 FITS frames (39 MB), K2-29220107062713.FITS → K2-29220107092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-220107.log (a189e4e0d8d7...), FinalLightCurve_K2-29 b_2022-01-06.png (4867067bb7d3...), FinalLightCurve_K2-29 b_2022-01-06.csv (8c64b9d18ea1...)

Compute resources

Wall time 102.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:05Z.